Cumulative Exam
Inorganic Chemistry
March 11, 2016

Instructions: Answer any five of the following ten questions (20 points each). Any answers provided beyond the requested five will not be graded.

1) Using a combination of diagrams and sentences, describe the band structure of p-doped and n-doped semi-conductors. Provide a diagram of the band structure for both p-doped and n-doped cases. Be sure to label the bands and draw / label the Fermi level in each case.

2) Using a combination of diagrams and sentences, describe the band structure of both an insulator and a conductor. Provide a diagram of the band structure for each case. Be sure to label the bands and draw / label the Fermi level in each case.

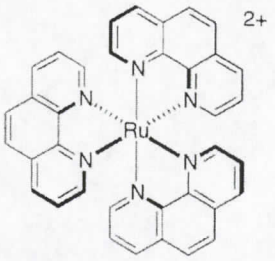
3) What determines the color of a semiconductor based Light Emitting Diode? Provide an answer that is more than a few words.

4) Describe how a photovoltaic device and a photodiode detector function. Include a discussion of the band structure of the devices. Include a diagram describing the band structure of each device and label the applied bias voltage in each case.

5) A Gouy balance is used to quantify the magnetic susceptibility of solids and materials. Describe how this device measures magnetic susceptibility. You may use any combination of written descriptions or diagrams in your answer.

6) What is Wilkinson's catalyst? Give an example of a reaction it catalyzes?

7) The "paddle-wheel" structure of $[Ru(phen)_3]^{2+}$ is drawn below. To what point group does this molecule belong?



8) Predict and draw the major product(s) of the reaction of Schwartz's reagent (zirconocene hydrochloride) and trans-2-pentene

9) Predict and draw the major product(s) of the reaction of Vaska's complex and methyl iodide

10) Provide a complete catalytic cycle and mechanism for the hydrosilylation of ethylene by diphenylsilane catalyzed by $Cp^*_2Y\text{-}CH(SiMe_3)_2$